

A Characterization Toolkit for Kurka's $c_0 \oplus X$ Borel Question

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Status

partial_result. The packet gives a Sobczyk–Kurka characterization, a quotient-persistence obstruction, an unconditional-embedding exclusion, a CDDK route dichotomy, a unique nonmeager class theorem, a paving-entry criterion, and a collapse-set obstruction.

This packet records a small toolkit for a question raised in Kurka's paper [1]: whether there exists a separable Banach space X such that the isomorphism class of $c_0 \oplus X$ is Borel. The first reduction uses Sobczyk's theorem to characterize the positive-example search: for separable X , the class $[c_0 \oplus X]$ is Borel if and only if X contains c_0 and $[X]$ is Borel. A quotient-persistence corollary then says that a positive example cannot split off c_0 with a quotient that is small in Kurka's sense; in particular, no positive example can embed into a space with an unconditional basis. The next reduction combines this with work of Cuth, Dolezal, Doucha, and Kurka to show that the Gurarii space is the unique nonmeager isomorphism class in their Polish norm codings, and hence to give a route dichotomy: a positive answer must either be the Gurarii Borel problem or a meager non- F_σ Borel class. Another corollary turns CDDK's theorem on spaces determined by pavings into a concrete positive-example entry point, while the unconditional-embedding exclusion rules out the naive c_0 -sum paving shortcut. The final reduction is the earlier collapse-set obstruction: any positive example must absorb a nonanalytic family of Kurka's finite-only Bourgain–Delbaen obstruction spaces.

Source Question

The source paper is Ondrej Kurka, *The isomorphism class of c_0 is not Borel*, arXiv:1711.03328 [1]. In the introduction, after Theorem 1.1, Kurka writes that his method proves non-Borelness of the isomorphism class of $c_0 \oplus X$ when X is not big in a sense, and then says that it is not known whether there is a space X for which the isomorphism class of $c_0 \oplus X$ is Borel.

only. Theorem 1.1 is an improvement indeed, since it is basically a statement about the structure of the class of $\mathcal{L}_\infty$ -spaces.

Actually, it follows from our method of proving Theorem 1.1 that the isomorphism class of $c_0 \oplus X$ is not Borel in the case that X is not big in a sense (in fact, we do not know if there is a space X such that the isomorphism class of $c_0 \oplus X$ is Borel).

Theorem 1.2. *Let X be a separable Banach space that satisfies any of the following conditions:*

- X does not contain an isomorphic copy of c_0 ,
- X does not contain an infinite-dimensional reflexive subspace,
- X is a subspace of a space with an unconditional basis.

Then the class of all Banach spaces isomorphic to $c_0 \oplus X$ is not Borel. Furthermore, the class of all Banach spaces isomorphic to a subspace of $c_0 \oplus X$ is not Borel as well.

Let us remark that the second part of this result has been already

Sobczyk–Kurka Characterization

The following reformulation is often the cleanest way to view Kurka’s question. It is not a full negative solution, because it transfers the remaining case to the general problem of finding Borel isomorphism classes among separable spaces which contain c_0 .

Theorem (Sobczyk–Kurka characterization). *Let X be a separable Banach space. Then the isomorphism class of $c_0 \oplus X$ is Borel if and only if X contains a copy of c_0 and the isomorphism class of X is Borel.*

Proof. If X contains no copy of c_0 , Kurka’s Theorem 1.2(i) says directly that the isomorphism class of $c_0 \oplus X$ is not Borel [1, Theorem 1.2]. Thus the equivalence is true in this case, since the right-hand side is false.

Assume now that X contains a subspace $E \simeq c_0$. By Sobczyk’s theorem [6], every isomorphic copy of c_0 in a separable Banach space is complemented. Hence $X = E \oplus Z$ for $Z = \ker P$, where $P : X \rightarrow E$ is a bounded projection. Therefore

$$c_0 \oplus X \simeq c_0 \oplus E \oplus Z \simeq c_0 \oplus c_0 \oplus Z \simeq c_0 \oplus Z \simeq X.$$

So, when X contains c_0 , the isomorphism classes $[c_0 \oplus X]$ and $[X]$ are the same subset of the standard Borel space of separable Banach spaces. Their Borelness is therefore equivalent. $\square$

Corollary (Equivalent positive-example problem). *Kurka’s question is equivalent to asking whether there exists a separable Banach space containing c_0 whose isomorphism class is Borel.*

Proof. If such a space Y exists, the theorem applied to $X = Y$ gives $[c_0 \oplus Y] = [Y]$, so Kurka’s question has a positive answer. Conversely, if $c_0 \oplus X$ has Borel isomorphism class for some separable X , the theorem implies that X contains c_0 and has Borel isomorphism class. $\square$

Corollary (Absorption form). *For a separable Banach space X , the following are equivalent:*

$$X \text{ contains } c_0, \quad X \text{ contains a complemented copy of } c_0, \quad X \simeq c_0 \oplus X.$$

Consequently, adding a c_0 -summand has only two descriptive-set-theoretic regimes: if X contains c_0 , then $[c_0 \oplus X] = [X]$; if X contains no c_0 , then $[c_0 \oplus X]$ is non-Borel by Kurka’s theorem.

Proof. The only nontrivial implication is that containing c_0 gives a complemented copy, which is Sobczyk's theorem in the separable setting. The decomposition $X \simeq c_0 \oplus Z$ then gives

$$c_0 \oplus X \simeq c_0 \oplus c_0 \oplus Z \simeq c_0 \oplus Z \simeq X.$$

The final sentence combines this absorption with Kurka's Theorem 1.2(i). $\square$

Corollary (Quotient-persistence obstruction). *Let Y be a separable Banach space containing c_0 . Suppose there is a subspace $E \simeq c_0$ such that the quotient Y/E satisfies at least one of the following conditions:*

1. Y/E contains no copy of c_0 ;
2. Y/E contains no infinite-dimensional reflexive subspace;
3. Y/E embeds into a Banach space with an unconditional basis.

Then the isomorphism class of Y is not Borel.

Consequently, if Y is a Borel positive example, then for every copy $E \simeq c_0$ in Y , the quotient Y/E must contain c_0 , must contain an infinite-dimensional reflexive subspace, and must fail to embed into any space with an unconditional basis.

Proof. By Sobczyk's theorem, the copy $E \simeq c_0$ is complemented in Y . Thus $Y \simeq E \oplus Z$, where Z is isomorphic to Y/E . Applying Kurka's Theorem 1.2 to $X = Z$ gives that the isomorphism class of $c_0 \oplus Z \simeq Y$ is not Borel whenever Z satisfies any of the three listed hypotheses. The final assertion is the contrapositive. $\square$

Corollary (Unconditional-embedding exclusion). *Let Y be a separable Banach space containing c_0 . If Y embeds into a Banach space with an unconditional basis, then the isomorphism class of Y is not Borel.*

In particular, a Borel positive example for Kurka's question cannot have an unconditional basis and cannot be isomorphic to a subspace of c_0 . Moreover no space of the form $(\bigoplus_n E_n)_{c_0}$, with E_n finite dimensional, can be a positive example.

Proof. Choose a copy $E \simeq c_0$ in Y . By Sobczyk's theorem, E is complemented, so $Y \simeq E \oplus Z \simeq c_0 \oplus Z$. If Y embeds into a Banach space with an unconditional basis, then the complemented subspace $Z \subseteq Y$ also embeds into such a space. Kurka's Theorem 1.2(iii) applied to $X = Z$ therefore gives that the isomorphism class of $c_0 \oplus Z \simeq Y$ is not Borel.

The first particular case follows by taking the ambient space to be Y itself when Y has an unconditional basis, or c_0 when Y is a subspace of c_0 . Finally, if each E_n is finite dimensional, then for each n one may choose a finite-dimensional almost-isometric embedding of E_n into some $\ell_\infty^{k_n}$, with distortion bounded uniformly, say by 2. Taking the c_0 -sum of these embeddings gives an isomorphic embedding

$$\left(\bigoplus_n E_n \right)_{c_0} \hookrightarrow \left(\bigoplus_n \ell_\infty^{k_n} \right)_{c_0} \simeq c_0.$$

Thus such a c_0 -sum is covered by the subspace-of- c_0 case. $\square$

CDDK Low-Complexity and Category Barrier

The next consequence is not a topology-free non-Borelness theorem. It is a statement in the Polish norm codings $\mathcal{B}$ and $\mathcal{P}_\infty$ used by Cuth, Dolezal, Doucha, and Kurka [7]. It nevertheless says that the easiest possible positive examples are ruled out.

Theorem (No simple positive example in the CDDK codings). *Let X be a separable Banach space and put $Y = c_0 \oplus X$. In the Polish spaces $\mathcal{B}$ and $\mathcal{P}_\infty$ of [7], the isomorphism class of Y is not an F_σ -set. Moreover, if the isomorphism class of Y is a G_δ -set, then Y is isomorphic to the Gurarii space $\mathbb{G}$.*

Consequently, a G_δ positive answer to Kurka's question is possible if and only if the isomorphism class of $\mathbb{G}$ is G_δ in these codings.

Proof. The space $Y = c_0 \oplus X$ contains c_0 , and hence is not reflexive. Therefore Y is not isomorphic to ℓ_2 . Cuth, Dolezal, Doucha, and Kurka prove that ℓ_2 is the unique separable infinite-dimensional Banach space whose isomorphism class is F_σ in $\mathcal{B}$ or $\mathcal{P}_\infty$ [7, Theorem 2.10]. Thus the isomorphism class of Y is not F_σ .

They also prove that, with the possible exception of spaces isomorphic to the Gurarii space, no isomorphism class is G_δ [7, Proposition 2.11]. Hence if the isomorphism class of Y is G_δ , then $Y \simeq \mathbb{G}$.

Finally, the Gurarii space is universal for separable Banach spaces; in particular it contains a copy of c_0 [8]. By Sobczyk's theorem, $\mathbb{G} \simeq c_0 \oplus Z$ for some Z , and therefore

$$c_0 \oplus \mathbb{G} \simeq c_0 \oplus c_0 \oplus Z \simeq c_0 \oplus Z \simeq \mathbb{G}.$$

Thus, if the isomorphism class of $\mathbb{G}$ is G_δ , taking $X = \mathbb{G}$ gives a G_δ positive example. The converse is the previous paragraph. $\square$

Corollary (The unique nonmeager isomorphism class). *In the CDDK Polish norm codings, the Gurarii space is the unique separable infinite-dimensional Banach space whose isomorphism class is nonmeager. Equivalently, every isomorphism class other than $[\mathbb{G}]$ is meager, while $[\mathbb{G}]$ is comeager.*

Proof. The isomorphism class of any fixed separable Banach space is analytic, hence has the Baire property. Suppose $[Y]$ is nonmeager. Then there is a nonempty open set U such that $[Y]$ is comeager in U . Cuth, Dolezal, Doucha, and Kurka prove that the isometry class of the Gurarii space is dense G_δ [7, Theorem 1.4(4)]. Therefore $[Y] \cap U$ and the isometry class of $\mathbb{G}$ are two comeager subsets of U , so they intersect. Thus Y is isomorphic to $\mathbb{G}$.

Conversely, $[\mathbb{G}]$ contains the dense G_δ isometry class of $\mathbb{G}$, so its complement is contained in a meager set. Hence $[\mathbb{G}]$ is comeager. $\square$

Corollary (Route dichotomy in the CDDK codings). *In the CDDK Polish norm codings, Kurka's question has a positive answer if and only if at least one of the following alternatives holds:*

1. *the isomorphism class of the Gurarii space $\mathbb{G}$ is Borel;*
2. *there is a separable Banach space Y containing c_0 such that $[Y]$ is Borel, meager, and not F_σ .*

Moreover, there is a nonmeager Borel positive example if and only if $[\mathbb{G}]$ is Borel.

Proof. If $[\mathbb{G}]$ is Borel, then $\mathbb{G}$ contains c_0 , so the Sobczyk–Kurka characterization gives a positive answer by taking $X = \mathbb{G}$. If the second alternative holds, the same characterization applied to $X = Y$ gives a positive answer.

Conversely, suppose Kurka's question has a positive answer. By the Sobczyk–Kurka characterization, there is a separable Y containing c_0 with $[Y]$ Borel. If $Y \simeq \mathbb{G}$, then $[\mathbb{G}]$ is Borel. If $Y \not\simeq \mathbb{G}$, the unique nonmeager class theorem gives that $[Y]$ is meager. Also $Y \not\simeq \ell_2$ because Y contains c_0 , and the no- F_σ part of the preceding theorem gives that $[Y]$ is not F_σ . This proves the dichotomy. The final assertion follows from the unique nonmeager class theorem. $\square$

Corollary (Paving entry point). *Let Y be a separable infinite-dimensional Banach space that contains c_0 . If Y is determined by its pavings, in the sense of Johnson–Lindenstrauss–Schechtman [3] as used by CDDK, then Kurka's question has a positive answer. The same conclusion holds if Y is determined by its finite-dimensional subspaces.*

If such a Y is not isomorphic to the Gurarii space, then its isomorphism class is meager and not F_σ in the CDDK codings.

Proof. Cuth, Dolezal, Doucha, and Kurka prove that a separable infinite-dimensional Banach space determined by its pavings has $G_{\delta\sigma}$, hence Borel, isomorphism class in $\mathcal{P}_\infty$ and in any admissible topology [7, the theorem on spaces determined by pavings]; intersecting with $\mathcal{B}$ gives the corresponding Borel statement there. Since Y contains c_0 , the Sobczyk–Kurka characterization gives a positive answer. A space determined by its finite-dimensional subspaces is determined by its pavings, as noted in CDDK, so the same proof applies. The last sentence is the route dichotomy and no- F_σ theorem above. $\square$

Candidate Audit from the Full-Result Push

The full-result push checked the most obvious positive examples against the literature. Kurka's introduction already records that the isomorphism classes of Pełczyński's universal space, $C(2^\mathbb{N})$, and $L_p[0, 1]$ for $1 < p < \infty$, $p \neq 2$, are not Borel [1]. By Milutin's theorem, $C[0, 1]$ is isomorphic to $C(2^\mathbb{N})$ [5], so the classical separable universal $C(K)$ -candidate is also ruled out.

On the positive side, Godefroy proved that the isomorphism classes of ℓ_p , $1 < p < \infty$, are Borel [4]. These spaces are reflexive and hence do not contain c_0 , so they cannot answer the reduced question. Kurka records that the Borel status of ℓ_1 was not known at the time [1]; in any case ℓ_1 has the Schur property and does not contain c_0 . Finally, the Gurarii space remains the main nonmeager candidate: it contains c_0 by universality [8], but CDDK explicitly state that they do not know whether its isomorphism class is Borel [7, Question 6.2].

Thus the standard shortcuts do not produce a full answer. In the CDDK codings, the remaining positive routes are exactly those in the route dichotomy: Gurarii Borelness, or a meager Borel non- F_σ c_0 -containing isomorphism class. The unconditional-embedding exclusion also rules out subspaces of c_0 , spaces with unconditional bases, and the natural c_0 -sums of finite-dimensional spaces as possible positive examples.

Notation from Kurka's Construction

Let $K(\mathcal{P}(\mathbb{N}))$ be the standard Borel space of compact families of subsets of $\mathbb{N}$, and let

$$\mathfrak{H} = \{\mathcal{M} \in K(\mathcal{P}(\mathbb{N})) : \mathcal{M} \text{ contains an infinite set}\}.$$

By Hurewicz's theorem, as recalled in [1, Theorem 4.3], $\mathfrak{H}$ is complete analytic and in particular is not Borel.

Kurka constructs a Borel map

$$\mathfrak{S} : K(\mathcal{P}(\mathbb{N})) \rightarrow \mathcal{SE}(C[0, 1])$$

such that $\mathfrak{S}(\mathcal{M})$ is isometric to a Bourgain–Delbaen $\mathcal{L}_\infty$ -space denoted here by $\mathfrak{X}_\mathcal{M}$. The construction has the following properties [1, Propositions 3.6 and 3.8, Lemma 4.1, and Theorem 4.3].

- If $\mathcal{M} \in \mathfrak{H}$, then $\mathfrak{X}_\mathcal{M} \simeq c_0$.
- If $\mathcal{M} \notin \mathfrak{H}$, equivalently if $\mathcal{M}$ consists only of finite sets, then $\mathfrak{X}_\mathcal{M}$ contains no copy of c_0 .
- There is a Borel set $\mathfrak{B} \subset K(\mathcal{P}(\mathbb{N}))$ such that $\mathfrak{H} \subset \mathfrak{B}$ and $\mathfrak{X}_\mathcal{M}^*$ is separable for every $\mathcal{M} \in \mathfrak{B}$.
- If $\mathcal{M} \notin \mathfrak{H}$ and $\mathfrak{X}_\mathcal{M}^*$ is separable, then every infinite-dimensional subspace of $\mathfrak{X}_\mathcal{M}$ contains an infinite-dimensional reflexive subspace.

For a fixed separable Banach space X , Kurka’s proof of Theorem 1.2 also uses a Borel map $\mathfrak{T}_X$ such that

$$\mathfrak{T}_X(\mathcal{M}) \simeq \mathfrak{X}_\mathcal{M} \oplus X.$$

Define the *collapse set*

$$\mathfrak{C}_X = \{\mathcal{M} \in \mathfrak{B} : \mathfrak{X}_\mathcal{M} \oplus X \simeq c_0 \oplus X\}.$$

Proof Intuition

Sobczyk’s theorem first removes a misleading feature of the direct sum: once X already contains c_0 , adding another c_0 -summand does not change the isomorphism class. Hence a positive example is exactly a Borel isomorphism class represented by a separable c_0 -containing space.

The CDDK input then separates the possible positive examples by topological size. A Borel positive example that is nonmeager is forced to be the Gurarii space. Therefore every non-Gurarii positive example has to be both meager and above the F_σ level. Conversely, a local finite-dimensional characterization such as being determined by pavings would automatically give Borelness; if one could find such a space containing c_0 , the Sobczyk–Kurka reduction would turn it into a positive answer. However, the unconditional-embedding exclusion shows that this cannot be the naive coordinate construction: no c_0 -sum of finite-dimensional blocks, and more generally no c_0 -containing subspace of an unconditional-basis space, can work.

The collapse-set argument gives a complementary obstruction. Kurka’s reduction always puts $\mathfrak{H}$ inside the collapse set, because $\mathfrak{X}_\mathcal{M} \simeq c_0$ for $\mathcal{M} \in \mathfrak{H}$. If the target isomorphism class $[c_0 \oplus X]$ were Borel, then $\mathfrak{C}_X$ would be Borel. A Borel set containing a complete analytic non-Borel set cannot have an analytic excess over it: if both $\mathfrak{H}$ and $\mathfrak{C}_X \setminus \mathfrak{H}$ were analytic, Lusin separation would make $\mathfrak{H}$ Borel. Thus Borelness of $[c_0 \oplus X]$ forces the finite-only collapse set to be nonanalytic.

Partial Theorem

Theorem (Collapse-set obstruction). *Let X be a separable Banach space and let $\mathfrak{C}_X$ be the collapse set above. If $\mathfrak{C}_X \setminus \mathfrak{H}$ is analytic, then the isomorphism class of $c_0 \oplus X$ is not Borel.*

Equivalently, if the isomorphism class of $c_0 \oplus X$ is Borel, then $\mathfrak{C}_X \setminus \mathfrak{H}$ is nonanalytic. In that case, for every $\mathcal{M} \in \mathfrak{C}_X \setminus \mathfrak{H}$, the space $c_0 \oplus X$ contains a complemented copy of $\mathfrak{X}_\mathcal{M}$; these $\mathfrak{X}_\mathcal{M}$ ’s contain no c_0 , have separable dual, and are reflexively saturated.

Proof. First note that $\mathfrak{H} \subset \mathfrak{C}_X$. Indeed, if $\mathcal{M} \in \mathfrak{H}$, then $\mathfrak{X}_{\mathcal{M}} \simeq c_0$, and therefore

$$\mathfrak{X}_{\mathcal{M}} \oplus X \simeq c_0 \oplus X.$$

Assume, toward the contrapositive, that the isomorphism class of $c_0 \oplus X$ is Borel. Since $\mathfrak{T}_X$ is Borel and $\mathfrak{B}$ is Borel, the set

$$\mathfrak{C}_X = \mathfrak{B} \cap \mathfrak{T}_X^{-1}(\{Y : Y \simeq c_0 \oplus X\})$$

is Borel in $K(\mathcal{P}(\mathbb{N}))$.

Suppose also that $\mathfrak{C}_X \setminus \mathfrak{H}$ is analytic. The two sets $\mathfrak{H}$ and $\mathfrak{C}_X \setminus \mathfrak{H}$ are disjoint analytic subsets of the standard Borel space $K(\mathcal{P}(\mathbb{N}))$. By Lusin's separation theorem, there is a Borel set D such that

$$\mathfrak{H} \subset D \quad \text{and} \quad D \cap (\mathfrak{C}_X \setminus \mathfrak{H}) = \emptyset.$$

Since $\mathfrak{H} \subset \mathfrak{C}_X$, we have

$$\mathfrak{C}_X = \mathfrak{H} \cup (\mathfrak{C}_X \setminus \mathfrak{H}),$$

and hence

$$\mathfrak{H} = D \cap \mathfrak{C}_X.$$

This makes $\mathfrak{H}$ Borel, contradicting Hurewicz's theorem. Therefore, if $\mathfrak{C}_X \setminus \mathfrak{H}$ is analytic, the isomorphism class of $c_0 \oplus X$ cannot be Borel.

The final assertion follows from the definition of the collapse set. If $\mathcal{M} \in \mathfrak{C}_X \setminus \mathfrak{H}$, then $\mathfrak{X}_{\mathcal{M}} \oplus X \simeq c_0 \oplus X$. The first summand $\mathfrak{X}_{\mathcal{M}}$ is complemented in $\mathfrak{X}_{\mathcal{M}} \oplus X$, so transporting the coordinate projection through an isomorphism gives a complemented copy of $\mathfrak{X}_{\mathcal{M}}$ inside $c_0 \oplus X$. Since $\mathcal{M} \in \mathfrak{B} \setminus \mathfrak{H}$, Kurka's properties listed above give that $\mathfrak{X}_{\mathcal{M}}$ contains no c_0 , has separable dual, and is reflexively saturated. $\square$

Corollary (Recovery of Kurka's obstruction pattern). *To prove that the isomorphism class of $c_0 \oplus X$ is not Borel, it is enough to prove that the finite-only collapse set $\mathfrak{C}_X \setminus \mathfrak{H}$ is analytic. In particular, Kurka's Theorem 1.2 proves non-Borelness by showing that this set is empty under each of the following hypotheses: X contains no c_0 , X contains no infinite-dimensional reflexive subspace, or X embeds into a space with an unconditional basis.*

Proof. The first sentence is exactly the theorem. Under each of Kurka's three hypotheses, the proof of [1, Theorem 1.2] shows that no finite-only parameter $\mathcal{M} \in \mathfrak{B} \setminus \mathfrak{H}$ can satisfy $\mathfrak{X}_{\mathcal{M}} \oplus X \simeq c_0 \oplus X$. Thus $\mathfrak{C}_X \setminus \mathfrak{H} = \emptyset$, which is analytic. $\square$

Limitations and Novelty Check

This is not a full solution of Kurka's question. The Sobczyk–Kurka characterization shows that a positive example is the same thing as a separable c_0 -containing Banach space with Borel isomorphism class, but it does not decide whether such a space exists. The CDDK barriers rule out F_σ examples, reduce the G_δ and nonmeager Borel cases to the Gurarii space, and leave only two possible positive routes in those codings: the Gurarii class is Borel, or there is a meager Borel non- F_σ c_0 -containing isomorphism class. The paving corollary is an entry criterion, not an existence theorem: the standard examples of paving-determined spaces mentioned by CDDK, such as certain ℓ_2 -sums of finite-dimensional spaces, are reflexive and therefore do not contain c_0 , while the tempting c_0 -sums of finite-dimensional spaces embed into c_0 and are excluded by the unconditional-embedding corollary. The collapse-set theorem does not rule out a space X for which the finite-only

collapse set is nonanalytic. Instead, it gives a necessary condition for any Borel positive example: $c_0 \oplus X$ must complementably absorb a nonanalytic parameter family of c_0 -free, separable-dual, reflexively saturated Bourgain–Delbaen spaces from Kurka’s construction.

A bounded local-index search found no previous packet for arXiv:1711.03328 or for the $c_0 \oplus X$ Borel question in this run before the present packet. A bounded web search for exact phrases such as “ $c_0 \oplus X$ Borel isomorphism class”, “isomorphism class of $c_0 \oplus X$ ”, and “Gurarii space isomorphism class Borel” did not reveal an explicit later solution. The 2022 paper of Cuth, Dolezal, Doucha, and Kurka records that even the Borel status of the Gurarii space’s isomorphism class is unknown [7, Question 6.2]. This supports the view that a complete negative solution to Kurka’s question would be a substantial result: by the characterization above, it would rule out every separable c_0 -containing space as a Borel-class example and, in particular, answer the Gurarii Borel question negatively. The results here are best viewed as derived characterizations and obstructions, not as a claimed complete answer.

Human Review Recommendation

Review should first check the Sobczyk absorption step: in the separable setting, if X contains c_0 , then $X \simeq c_0 \oplus Z$ and consequently $c_0 \oplus X \simeq X$. Review should also check the descriptive-set-theoretic step: if $\mathfrak{C}_X$ is Borel, $\mathfrak{H} \subset \mathfrak{C}_X$, and $\mathfrak{C}_X \setminus \mathfrak{H}$ is analytic, then Lusin separation indeed implies $\mathfrak{H}$ is Borel. Finally, review should check that Kurka’s Borel selector $\mathfrak{T}_X$ is sufficient to make $\mathfrak{C}_X$ Borel when the target isomorphism class is Borel. For the CDDK barriers, review should check that the F_σ , G_δ , and meagerness assertions are understood in the CDDK Polish norm codings rather than as topology-free statements about the Effros Borel structure. The final polish corollaries should be checked as reusable characterizations: the route dichotomy should be equivalent to the Baire-category dichotomy plus Sobczyk–Kurka, and the paving entry point should use only CDDK’s $G_{\delta\sigma}$ theorem for spaces determined by pavings together with c_0 -absorption. The unconditional-embedding exclusion should be checked as an immediate application of Kurka’s Theorem 1.2(iii) to the Sobczyk complement.

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